

Full length article

Biomechanical characteristics of handstand walking initiation

Michal Grabowiecki^{a,b}, Lorenzo Rum^a, Luca Laudani^b, Giuseppe Vannozzi^{a,*}^a Department of Movement, Human and Health Sciences, University of Rome “Foro Italico”, Rome, Italy^b Cardiff School of Sport and Health Sciences, Cardiff Metropolitan University, Cyncoed Campus, Cyncoed Road, CF23 6XD, Cardiff, UK

ARTICLE INFO

Keywords:

Gymnastics

Balance

Centre of pressure

Biomechanics

Anticipatory postural adjustments

ABSTRACT

Background: The initiation in human locomotion is defined as the transition between upright stance and steady-state gait. While past literature abundantly investigated the initiation in bipedal gait, the initiation of handstand walking remains unexplored.

Research question: The current study aims to characterise the centre of pressure (CoP) and centre of mass (CoM) trajectory of handstand walking initiation as well as the spatiotemporal and kinematic parameters and balance strategy of this task. Also, the study examined the CoP trajectory similarity within- and between-participants using a coefficient of multiple correlation analysis.

Methods: Nineteen gymnasts took part in this study. Handstand walking initiation trials were recorded using force plates and a stereophotogrammetric system. CoM and CoP trajectories were analysed during the Baseline, Preparation and Execution phases of the motor task.

Results: We found that to successfully perform the handstand walking initiation, a shift of the CoM forward and towards the stance hand is required as a result of a lateral and posterior CoP shift. All participants performed a similar CoP pattern in the mediolateral direction, whereas two anteroposterior CoP displacement strategies were identified across participants based on different timing execution of posterior CoP shift. While CoP and CoM kinematic differences were identified during the Preparation Phase due to the adopted strategy, no significant difference was found in the Execution Phase for the spatiotemporal and kinematic characteristics.

Significance: A better understanding of the required CoP/CoM patterns and balance control provides the basis for further neuromechanics research on the topic and could contribute to individualise training protocols to improve the learning of the task.

1. Introduction

The transition task between standing and walking, namely gait initiation, has been widely investigated in the literature [1,2]. A review by Yiou et al. [3] shows that this transition begins with individuals performing anticipatory postural adjustments (APAs) that occur before the onset of the first step. In gait initiation, APAs involve a coordinated muscle activation sequence which shifts the centre of pressure (CoP) posteriorly and towards the swing limb, resulting in a movement of the centre of mass (CoM) anteriorly and towards the stance limb. Subsequently, the CoP moves towards the stance side and toe-off begins with anterior and medial CoP shift [3–5]. This CoP and CoM pattern creates a suitable condition for the body to be propelled forward while maintaining the balance control during the first step execution [6–8].

In recent decades, there has been a growing interest to uncover the postural strategies aimed at maintaining a handstand position,

particularly in athletes [9–14]. All artistic gymnasts either perform handstand balance or move from a handstand position using all artistic gymnastic equipment [15]. The CoP pattern during handstand balance resembles the CoP pattern in normal upright posture [9]. Kerwin and Trewartha [10] also reported that upper limb kinematics in a handstand (involving the shoulder, elbow, and wrist joints) reflected lower limb kinematics in the upright posture (involving hip, knee and ankle joints). Despite these similarities, handstand balance is a continuous dynamic condition where greater anteroposterior CoP oscillations occur compared to those in the mediolateral direction [11,13]. In addition, this anteroposterior CoP oscillation is also greater than that observed during bipedal stance [12].

Handstand walking is one of the conditions that is often used as a skill progression to challenge gymnasts' balance ability [15] and even as a strength exercise in one of the rehabilitation stages after an arm injury of gymnasts [16]. However, Earhart and Mosiman [14] are the only

* Corresponding author at: Department of Movement, Human and Health Sciences, University of Rome “Foro Italico”, Piazza Lauro de Bosis 6, 00135, Roma, Italy.
E-mail address: giuseppe.vannozzi@uniroma4.it (G. Vannozzi).

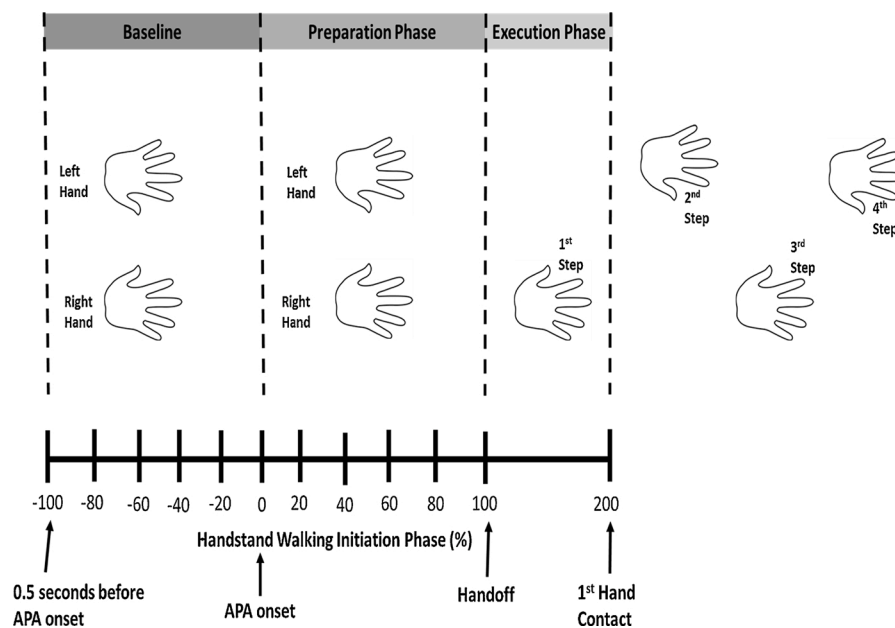


Fig. 1. Graphical representation of the task and definition of the key phases.

authors to date that analysed handstand walking and focused on comparing it with bipedal walking. Interestingly, they found that individuals used nearly identical cadences for handstand walking and bipedal walking, while shorter strides, a wider base of support, and longer double support time were found in handstand walking. The transition from a stable handstand balance to continuous handstand walking could play a fundamental role in the success of the task and it can be referred to as handstand walking initiation. To the best of the authors' knowledge, the mechanics of handstand walking initiation has not been previously investigated.

Therefore, this study aims to characterise APAs and balance control using CoP and CoM patterns, and the spatiotemporal and kinematic parameters during handstand walking initiation. Based on the previously reported similarities between handstand balance and upright posture, it was hypothesised that CoP would show a preparatory posterior and lateral shift towards the stepping hand before the first hand movement. As a consequence, the whole-body CoM will be propelled forward and laterally towards the stance hand supporting the initiation of handstand walking. In light of the differences between anteroposterior and mediolateral CoP as well as the dynamic nature of handstand balance, our second hypothesis is that the anteroposterior CoP during handstand walking initiation would exhibit greater variability compared to the mediolateral CoP.

2. Methods

2.1. Participants

Nineteen gymnasts (age: 23.2 ± 4.6 years; body mass: 60.5 ± 9.6 kg; height: 1.67 ± 0.07 m; gymnastic experience: 14.4 ± 4.7 years) volunteered to take part in this study. All participants were free from neurological and musculoskeletal disorders affecting the upper limbs. Inclusion criteria were a minimum of 5 years of gymnastic experience, being able to hold a stable handstand position for 10 s and being able to take four controlled handstand steps. Written informed consent was obtained before inclusion in the study. The procedures were approved by the institutional review board of the University of Rome "Foro Italico".

2.2. Protocol

All participants performed a 10–15 min warm-up which involved 3 sets of 5 s handstand hold, 3 sets of 4 handstand steps and any additional stretches or progressions freely chosen by the individual. Each participant performed five handstand walking initiation trials. In the current study, a handstand step is referred to as picking up movement of one hand and placing it back down on the ground in the forward direction. Participants were instructed to perform a stable handstand position for at least three seconds before initiating a handstand step with their dominant hand at a natural speed. Participants were required to perform at least four consecutive handstand steps, maintaining the same initial hand position for all trials throughout the experimental session. A three-minute rest was given between each trial. The whole trial was considered valid for analysis when a stable handstand position was held in a straight line with legs together and four successful steps were performed [15]. To minimise the effect of fatigue, the researcher helped to establish a stable handstand position before the three-second handstand hold.

2.3. Kinetic and kinematic data

Kinetic data acquisition was performed with two piezoresistive force plates (40×40 cm Bertec Corp, Columbus, Ohio, USA) positioned side by side at ground level so that the gymnast was able to place one hand on each force plate. CoP and ground reaction force (GRF) data were collected at a sampling rate of 1000 Hz, recorded on a PC using Nexus Software (Vicon Motion Systems, Oxford, UK). The Winter et al. [17] method was used to compute the net CoP from each of the force plates. A seven-camera stereophotogrammetric system (Vicon MX3 Oxford, UK) was used to reconstruct the 3D position of 39 retro-reflective spherical markers located on the participants' skin to create a 15-body segment 3D model (Vicon Plug-in-gait full-body model) [18,19]. Kinematic data were collected at 100 Hz. Kinetic and kinematic data were filtered using a second-order low-pass Butterworth filter with a cut-off frequency of 15 Hz [20].

2.4. Data analysis

Customised event detection algorithms and interactive graphical analysis programs (MATLAB R2020a; MathWorks, Natick, MA) were

Table 1

Definition of CoP, CoM, Phases, Velocity, Interval parameters and group definition. The following parameters are proposed consistently with those typically used in characterising bipedal gait initiation [3]. A subdivision of the Baseline and Preparation phases into 25 % intervals of the relevant duration was performed to better understand the biomechanical behaviour during these phases.

Time events and spatial characteristics	APA onset	Due to a large AP CoP oscillation in handstand balance [12], APA onset was defined as the time instant in which the mediolateral CoP velocity exceeded by 3 S.D. the mean value calculated over a 0.5-second interval chosen visually by the researcher during the period before any clear mediolateral CoP displacement occurred [21]
	Baseline	The 0.5-second interval before the APA onset (Fig. 1)
	APA Phase	The period from APA onset to maximal mediolateral CoP displacement
	Preparation Phase	The period from APA onset to the first Hand-off (Fig. 1)
	Execution Phase	The period between first Hand-off and first Hand-contact (Fig. 1)
	Hand-off	The moment when the GRF registered under the stepping hand reached 0 N
	Hand-contact	The lowest vertical position of finger marker after the Hand-off
CoP	25 % intervals	Baseline and Preparation Phase were divided into 25 % intervals where anteroposterior CoP velocity, ankle velocity and hip angle velocity were averaged
	APA onset Position	CoP at APA onset in mediolateral and anteroposterior direction was expressed as a percentage of the BoS boundaries
	AP CoP Displacement	Maximal posterior and maximal anterior CoP displacement were calculated during the Preparation Phase
	ML CoP Displacement	Maximal mediolateral CoP displacement towards the stepping hand was calculated during the Preparation Phase
CoM	APA time	Time of APA Phase
	APA onset Position	CoM at APA onset in mediolateral and anteroposterior direction was expressed as a percentage of the BoS boundaries
	AP/ML CoM Hand-off and Hand-contact	Anteroposterior and mediolateral CoM displacement at Hand-off and Hand-contact were calculated with respect to CoM position at APA onset
Velocity Parameters	AP CoP Velocity	Anteroposterior CoP velocity was calculated using the first derivative of anteroposterior CoP displacement during: Baseline, APA onset, Preparation Phase, Hand-off
	AP CoM Velocity	Anteroposterior CoP velocity was calculated using the first derivative of anteroposterior CoP displacement at the instant of APA onset, Hand-off and Hand-contact
	AP Ankle Velocity	Anteroposterior ankle velocity was calculated using the first derivative of mean anteroposterior left and right ankle marker displacement, during: Baseline, APA onset, Preparation Phase and Hand-off
	Hip Angle Velocity	Hip angle velocity was calculated using the first derivative of mean left and right hip angle displacement in the sagittal plane during the Baseline, APA onset, Preparation Phase and Hand-off
Step Characteristics	Step length Step width	Step length and width were defined as anteroposterior and mediolateral distance between finger markers of right- and left-hand during Hand-contact, respectively
	Step Time	Time of Execution Phase
	Step Velocity	Step length / Step time
	Step Phase%	Percentage of time spent during the Execution Phase in relation to the whole handstand walking initiation duration

used to analyse the data. This provided a phase analysis for the task (see Fig. 1) and a list of the defined events and proposed parameters (Table 1). The base of support (BoS) was estimated using the finger and wrist markers' coordinates.

Three trials that resembled the closest visual and Coefficient of Multiple Correlation (CMC) similarities in both anteroposterior and mediolateral CoP trajectories were selected for further analysis. Following the inspection of anteroposterior (AP) CoP trajectories, two groups were identified based on the clear adoption of posterior shift (PSG) or anterior shift (ASG) strategy during the Preparation Phase. CMC analysis also assessed the similarity of CoP trajectories within (Intra-subject) and between (Inter-subject) participants during Baseline, Preparation and Execution Phases (Fig. 1) [22,23].

Table 2

Intra-subject and inter-subject CMC output for AP and ML CoP trajectories during Baseline, Preparation and Execution Phase (Mean $\pm$ SD). CMC value closer to 1 indicates high similarity of trajectories, CMC value closer to 0 indicates low similarity of trajectories. CMC value of >0.85 represents high similarity and is highlighted in bold [23].

Whole Group n = 19	Baseline		Preparation Phase		Execution Phase	
	AP	ML	AP	ML	AP	ML
Intra-subject	0.47 $\pm$ 0.17	0.43 $\pm$ 0.18	0.67 $\pm$ 0.24	0.94 $\pm$ 0.05	0.53 $\pm$ 0.16	0.60 $\pm$ 0.11
Inter-subject	0.08	0.11	0.09	0.90	0.12	0.13
PSG n = 12						
Intra-subject	0.44 $\pm$ 0.18	0.41 $\pm$ 0.20	0.72 $\pm$ 0.20	0.93 $\pm$ 0.05	0.54 $\pm$ 0.18	0.61 $\pm$ 0.10
Inter-subject	0.10	0.14	0.27	0.89	0.15	0.16
ASG n = 7						
Intra-subject	0.52 $\pm$ 0.16	0.48 $\pm$ 0.13	0.53 $\pm$ 0.30	0.96 $\pm$ 0.05	0.45 $\pm$ 0.20	0.66 $\pm$ 0.04
Inter-subject	0.41	0.21	0.42	0.90	0.21	0.22

2.5. Statistical analysis

The normal distribution of data was tested using a Shapiro-Wilk test and the presence of outliers was checked through the Z-score method. Differences between groups were assessed using Mann-Whitney U tests for not normally distributed spatiotemporal and kinematic parameters, while an independent T-test was considered otherwise. The effect size was calculated using Cohen's d and r-squared, with a value of 0.20 representing a small, 0.50 a moderate, and 0.80 a large effect [24]. To understand the differences in sagittal plane behaviour, a comparison of AP CoP velocity, ankle marker velocity and hip joint angular velocity between the PSG and ASG was performed. Holm-Bonferroni corrections were performed on Baseline intervals of 25–50 %, 50–75 %, 75–100 %, APA onset and Preparation Phase intervals of 0–25 % and 25–50 %. The remaining intervals were not analysed as the numerical difference between groups became lower and similarity in behaviour was present. To understand the relationship between spatiotemporal and kinematic parameters of the task, Pearson's multiple correlation test was used. The significance level α was set to 0.05 and all statistical tests were performed using the SPSS 23.0 software (Chicago, IL, USA).

3. Results

3.1. Curve similarity

Intra-subject and inter-subject CMC results are summarised in Table 2. The Whole Group, PSG and ASG intra-subject CMC were greater than inter-subject CMC in each of the three phases. During the Preparation Phase, the Whole group, PSG and ASG achieved lower intra- and inter-subject CMC scores for the AP CoP compared to the ML CoP. The intra-subject CMC score for the AP CoP during the Preparation Phase was higher for PSG compared to the ASG and the Whole group CMC scores.

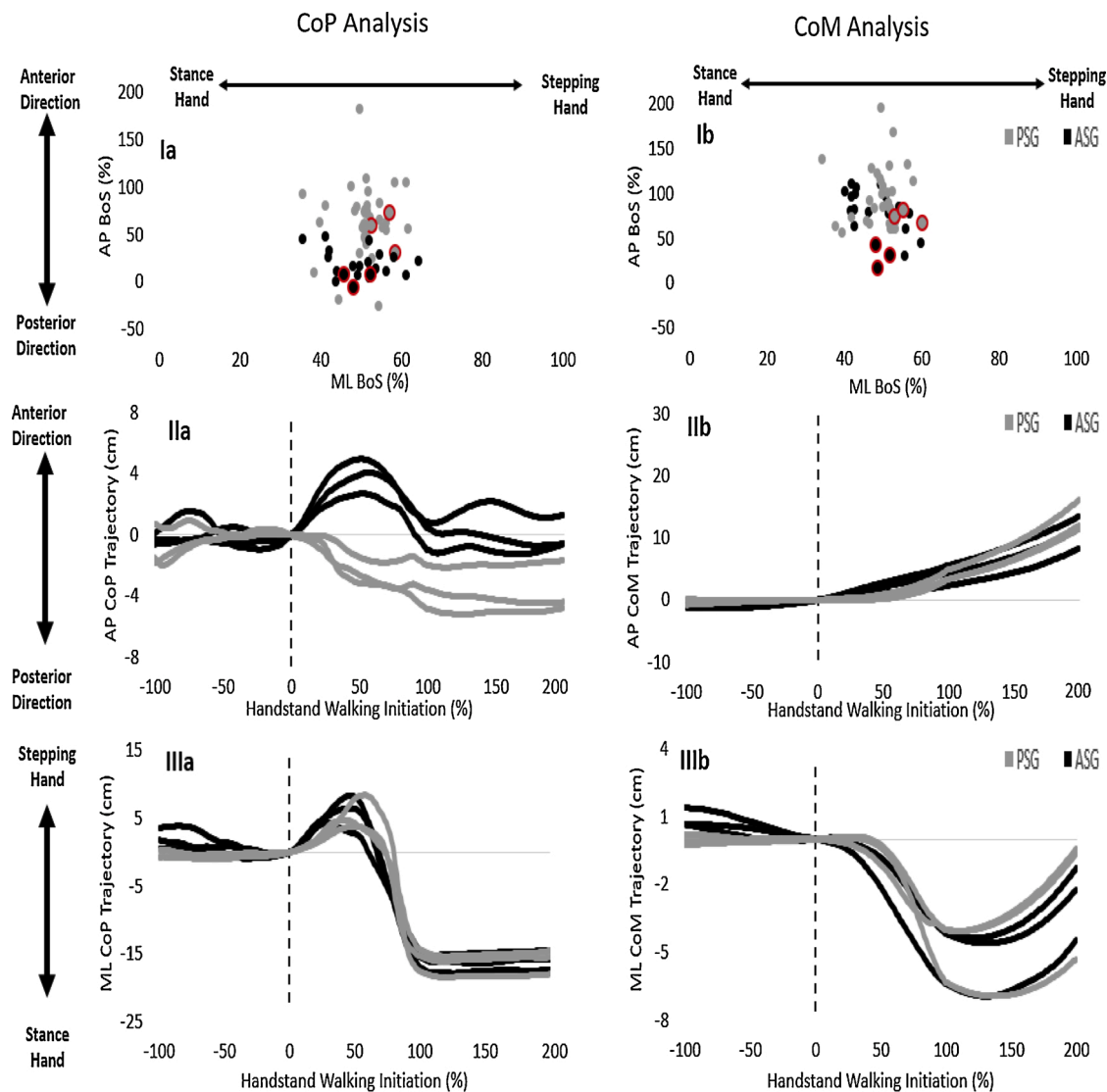


Fig. 2. CoP (Ia) and CoM (Ib) initial position at APA onset presented as percentage of BoS for PSG (Grey dots) and ASG (Black dots). In the horizontal axis 0% indicates the CoP and CoM towards the stance hand and 100 % indicates the stepping hand. In the vertical axis 0% indicates the CoP and CoM towards the wrist markers and 100 % indicates the finger markers. Points with red outline represents trials demonstrated in IIa, IIIa, IIb and IIIb. CoP (IIa and IIIa) and CoM (IIb and IIIb) trajectory during Baseline (-100-0%), Preparation (0-100 %) and Execution (100-200 %) Phase represented by one participant from posterior shift group (PSG - grey line) and anterior shift group (ASG - black line). In order to represent the CoP and CoM trajectories of the task, one participant from each group was chosen as a good representative of the general CoP and CoM behaviour for the group. These two participants also achieved an intra-subject CMC score among the highest in their group. Both CoP and CoM trajectories during all three phases were aligned to the position at APA onset time instant (vertical dashed line) (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.).

3.2. CoP and CoM analysis

A significant difference between groups was observed in the AP CoP position at APA onset with respect to the percentage of BoS (PSG: 62.2 ± 37.7 ; ASG: 18.2 ± 14.6 ; $T(17) = 3.49$, $p = 0.003$, $d = 1.67$), while no difference was observed in the ML CoP position and both of the CoM positions (see Fig. 2 Ia and Ib, respectively).

Subfigures 2IIa and 2IIIa illustrate the CoP trajectories from two representative participants of the ASG and PSG groups. Both participants showed varying ML and AP CoP trajectories during the Baseline. During the Preparation Phase, the ASG participant exhibited an anterior CoP shift followed by a posterior CoP shift, whereas, the PSG participant only exhibited a posterior CoP shift. In the ML direction, both participants shifted their CoP towards the stepping hand and, then, towards the stance hand. During the Execution Phase, the AP and ML CoP plateaued.

Anterior CoM shift was observed earlier in the ASG compared to the PSG during the Baseline Phase (Fig. 2IIb). Both groups shifted their CoM

in the anterior direction during the Preparation and Execution Phase. Small CoM movements in the mediolateral direction can be observed by both groups during the Baseline Phase. Both groups shifted their CoM laterally towards the stance hand at the beginning of the Preparation Phase. During the Execution Phase, the CoM for both groups shifted medially towards the stepping hand.

3.3. Spatiotemporal and kinematic parameters

Table 3 outlines the differences of the spatiotemporal and kinematic parameters between both groups. The PSG showed greater posterior CoP displacement during the Preparation Phase compared to the ASG ($T(13.3) = -7.38$, $p < 0.01$, $d = 9.4$). Greater anterior CoP displacement was achieved by the ASG during the Preparation Phase compared to the PSG ($U = 9$, $Z = -2.79$, $p < 0.01$, $r^2 = 0.41$). In addition, AP CoM velocity at APA onset was significantly higher in the ASG than the PSG ($T(17) = -2.43$, $p = 0.026$, $d = 3.47$). No significant difference was found in AP CoM

Table 3

Spatiotemporal and kinematic parameters from PSG and ASG (Mean $\pm$ SD). A positive value for AP CoP/CoM corresponds to the movement towards the anterior direction. A positive value for ML CoP/CoM corresponds to the movement towards the stepping hand direction. Significant difference between groups are highlighted in bold and are indicated with * symbol ($p < 0.05$).

Variables		PSG	ASG
CoP Parameters	Posterior CoP Displacement	$-2.33 \pm$	$-0.21 \pm$
	Preparation Phase (cm)	0.94	0.24*
	Anterior CoP Displacement	0.64 $\pm$	2.39 $\pm$
	Preparation Phase (cm)	0.66	1.03*
	ML CoP Displacement (cm)	7.14 ± 4.51	5.90 ± 3.12
	APA Time (s)	$199.97 \pm$	$201.62 \pm$
		44.61	61.53
	Ap CoM Displacement at Hand-off (cm)	4.10 ± 1.60	3.96 ± 1.24
	Ap CoM Displacement at Hand-contact (cm)	$11.90 \pm$	$11.30 \pm$
		3.27	2.66
CoM Parameters	ML CoM Displacement at Hand-off (cm)	$-4.63 \pm$	$-4.38 \pm$
		2.17	1.66
	ML CoM Displacement at Hand-contact (cm)	$-2.06 \pm$	$-0.81 \pm$
		2.39	1.94
	AP CoM Velocity at APAOnset (cm/s)	2.67 $\pm$	4.95 $\pm$
		1.78	2.27*
	AP CoM Velocity at Hand-off (cm/s)	$18.62 \pm$	$15.72 \pm$
		7.09	4.56
	AP CoM Velocity at Hand-contact (cm/s)	$46.06 \pm$	$40.49 \pm$
		9.25	7.87
Step Parameters	Step Length (cm)	$30.47 \pm$	$29.25 \pm$
		7.56	5.70
	Step Width (cm)	$45.27 \pm$	$46.43 \pm$
		5.16	7.57
	Step Time (s)	0.25 ± 0.05	0.28 ± 0.03
	Step Velocity (cm/s)	$123.17 \pm$	$105.56 \pm$
Global Parameters		34.52	24.78
	Step Phase (%)	$34.77 \pm$	$37.76 \pm$
		4.66	6.65
	Prep Duration (s)	0.48 ± 0.06	0.49 ± 0.13
	Gait Duration (s)	0.73 ± 0.09	0.77 ± 0.13

at hand-off ($p > 0.05$) or in the rest of the spatiotemporal and kinematic parameters between the two groups ($p > 0.05$).

3.4. Phase intervals

Fig. 3 (A–C) illustrates the differences in sagittal plane behaviour between both groups across all three phases. Significantly greater posterior CoP velocity was observed at 50–75 % of Baseline in the ASG compared to the PSG ($T(17) = 3.02$, $p < 0.01$, $d=4.39$). Significantly greater posterior CoP velocity was also observed at the APA onset, 0–25 % and 25–50 % intervals of Preparation Phase in the PSG compared to the ASG ($U=7$, $Z=-2.96$, $p < 0.01$, $r^2 = 0.46$; $T(8.50)=-5.17$, $p < 0.001$, $d=8.14$; and $T(17)=-3.83$, $p = 0.001$, $d=5.81$, respectively).

Significantly greater posterior ankle velocity was found in the ASG than in the PSG during the 25–50 % and the 50–75 % of the Baseline intervals ($T(17) = 4.06$, $p < 0.01$, $d=6.50$ and $T(17)=3.28$, $p < 0.01$, $d=5.19$ respectively). The hip angular velocity showed significantly greater flexion velocity during the 25–50 % and 50–75 % Baseline intervals in the ASG than in the PSG ($T(17)=-3.26$, $p < 0.01$, $d=4.82$ and $U=9$, $Z=-2.78$, $p < 0.01$, $r^2 = 0.41$, respectively).

3.5. Correlations

The AP CoP velocity was positively correlated with the ankle velocity during the 25–50 % interval of Baseline in the PSG ($r = 0.522$, $p < 0.01$) and the ASG ($r=0.623$, $p < 0.01$). Step length was positively correlated in both the PSG and the ASG with AP CoM velocity at APA onset ($r=0.419$, $p = 0.01$; $r=0.646$, $p < 0.01$, respectively), AP CoM velocity at hand-off ($r=0.694$, $p < 0.01$; $r=0.802$, $p < 0.01$, respectively), AP CoM velocity at hand-contact ($r=0.512$, $p < 0.01$; $r=0.856$, $p < 0.01$,

respectively), and AP CoM displacement at hand-contact ($r=0.848$, $p < 0.01$; $r=0.789$, $p < 0.01$, respectively). Step width was positively correlated with ML CoP displacement in the PSG ($r=0.580$, $p < 0.01$) and the ASG ($r=0.436$, $p = 0.022$). However, only the PSG increased their step width with greater ML CoM displacement towards the stance hand at hand-off ($r=0.501$, $p < 0.01$). In addition, the ML CoP displacement was negatively correlated with ML CoM displacement at hand-off in the PSG only ($r= 0.579$, $p < 0.01$).

4. Discussion

In the present study, for the first time to the authors' knowledge, the CoP and CoM trajectory patterns, balance control, spatiotemporal and kinematic parameters were analysed and characterised in handstand walking initiation. Partly in accordance with our first hypothesis, the main result of the study was that two different CoP strategies were identified during the Preparation Phase, with one of them resembling the strategy previously reported in bipedal gait initiation [1–5]. Despite these different and peculiar preparation strategies, only one main biomechanical pattern was observed in all participants during the Execution Phase of the first hand movement. In accordance with our second hypothesis, the CMC analysis demonstrated that the anteroposterior CoP exhibits greater variability compared to the medio-lateral CoP; this is also highlighted by continuous adaptations of the CoP trajectory along the sagittal direction.

Based on the bipedal gait initiation literature, the postural adjustments that help in achieving the optimal balance and propulsive requirements for step execution are performed in the Preparation Phase [1–5]. In this study, the postural strategy implemented by the PSG involved a posterior CoP shift during the Preparation Phase, thereby resembling what is commonly reported in bipedal gait initiation literature. On the other hand, the ASG showed a posterior CoP shift as early as the Baseline Phase, presenting a different postural strategy. The reason why the ASG did not perform the necessary posterior CoP shift during the Preparation Phase was due to a sufficient posterior CoP shift during the Baseline phase where the above mentioned continuous CoP movements take place. Consequently, this posterior CoP shift during the Baseline in the ASG allowed the participants to achieve greater forward CoM velocity at APA onset compared to the PSG. However, it is unclear why the ASG anticipated the posterior CoP shift during the Baseline. We hypothesise that, from a biomechanical standpoint, the ASG achieved the forward CoM velocity needed to initiate the first step at APA onset, thus no further CoP posterior shift during the Preparatory Phase was necessary due to earlier body adjustments. In support of this hypothesis, the ASG performed a greater hip angular velocity and lower ankle marker velocity during the 25–50 % and 50–75 % Baseline compared to the PSG. The positive correlation between ankle marker velocity and CoP velocity at 25–50 % of the Baseline suggests that this movement of the lower body in the posterior direction could be related to the early posterior CoP shift in the ASG.

Based on the results of anteroposterior initial position of the CoP at APA onset expressed as a percentage of the BoS, we propose that the anterior shift strategy is implemented when the CoP is closer to the posterior boundary of the BoS and behind the vertical projection of the CoM at the onset of handstand walking initiation. On the contrary, the posterior shift strategy is performed when the CoP position is beyond the CoM vertical projection and closer to the anterior boundary of the BoS. In these two strategies, the CoP is therefore shifted to decelerate (anterior shift strategy) or propel (posterior shift strategy) the CoM motion in the forward direction according to the initial handstand balance conditions. These findings can be further supported by previous works which demonstrated that gymnasts constantly shift their anteroposterior CoP during stable handstand balance [9–13]. Therefore, we propose that the adoption of the different strategy is determined by the position of the anteroposterior CoP in relation to the vertical projection of the CoM at the moment of deciding to begin the lateral thrust during

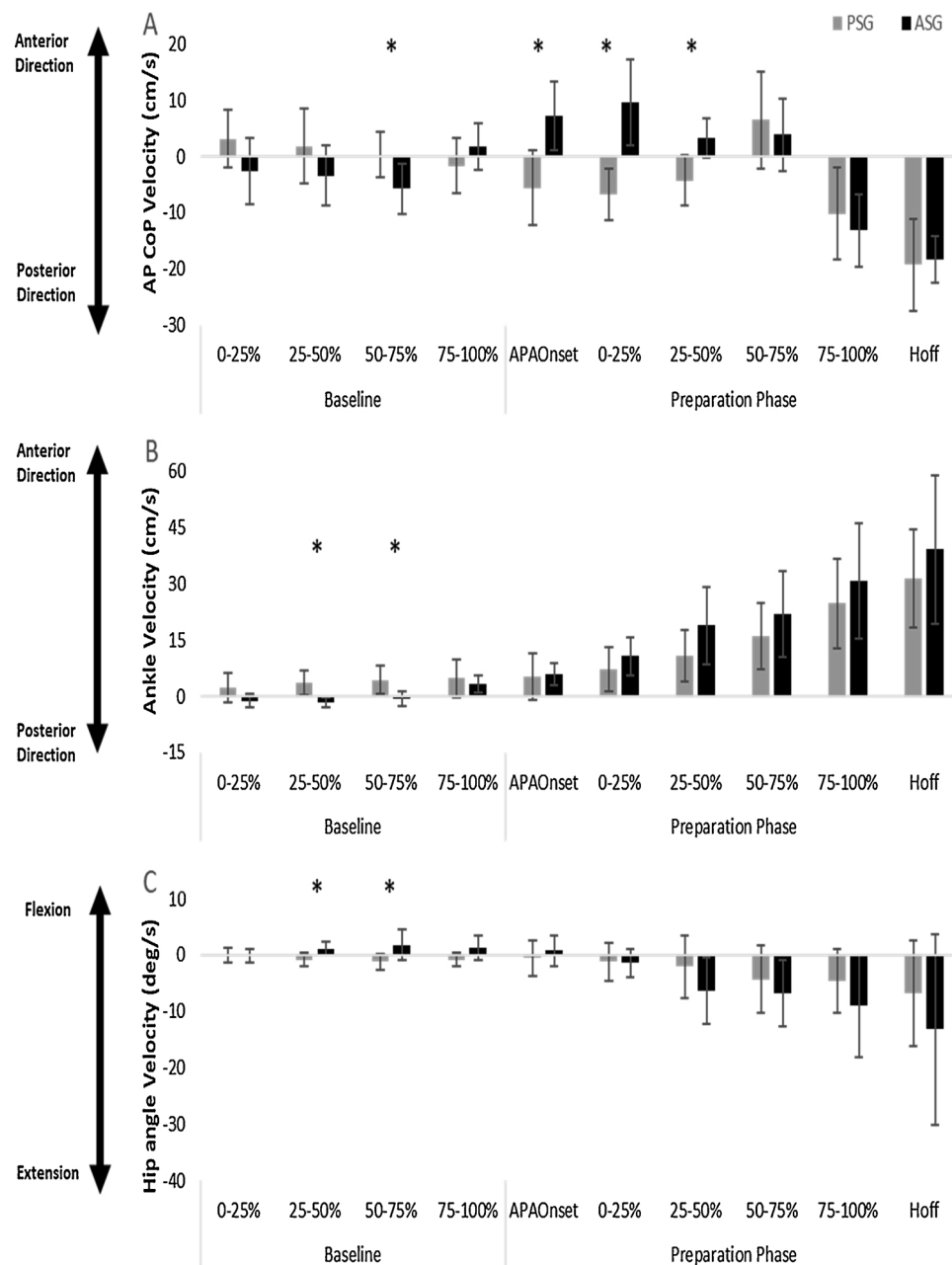


Fig. 3. Mean $\pm$ SD of PSG (grey) and ASG (black) AP CoP velocity (A), AP ankle velocity (B) and hip angle velocity (C). * Indicates a significant difference between both groups ($p < 0.05$).

handstand walking initiation.

The CMC analysis demonstrates greater variability in anteroposterior CoP trajectory behaviour between and within groups during the Preparation Phase compared to the mediolateral CoP trajectory behaviour. These results are in accordance with previously reported studies analysing handstand balance, where greater changes in the anteroposterior CoP behaviour compared to the mediolateral CoP behaviour were reported [11,13]. Moreover, Slobounov and Newell [12] confirmed bigger deviations in the sagittal plane in handstand balance compared to an upright position. Smaller base of support, higher vertical position of CoM and weaker support limb muscles are the biomechanical constraints that differentiate handstand balance to an upright stance. These are also present during the handstand walking initiation. These biomechanical constraints could explain the CoP and CoM patterns reported in the current study, especially the variability in the sagittal plane kinematics. However, more work needs to be done to fully understand the

effect of these biomechanical constraints on handstand walking initiation.

Despite the greater anteroposterior CoP behaviour variability, the mediolateral CoP behaviour was kept similar across individuals. This could suggest that the mediolateral CoP behaviour is required to be performed in a unique manner so that the balance control throughout the handstand walking initiation is maintained. In the current study, the control of mediolateral balance resembled the balance control previously reported in bipedal gait initiation [1–5]. During the initial phase of handstand walking initiation, participants shifted their CoP laterally towards the stepping hand to move the whole-body CoM in the opposite direction, towards the stance hand. This repositioning of the CoM before hand-off provides the necessary condition to maintain balance during the change in the base of support (from double- to single-handstand) that results from lifting and moving forward the stepping hand. Based on a mathematical model of the body falling freely under the action of

gravity [6], an increase in mediolateral CoM displacement towards the stance hand helps in avoiding an excessive fall of CoM toward the stepping hand side during the step execution. Subsequently, placing the hand on the ground allowed participants to break the lateral body fall, with stepping characteristics being related to the previously performed preparatory adjustments. Since an increase in mediolateral CoP displacement towards the stepping hand results in an increase in mediolateral CoM displacement towards the stance hand at handoff, it is reasonable to propose that the positive correlation between the mediolateral CoP displacement and step width in both groups could indicate an attempt to preserve balance and avoid falling; attempting to increase the CoM distance away from the stepping hand i.e. the side that the body is falling towards during the step execution, could lead to greater stability. Therefore, during the stable step execution, participants can place the hand in the most effective position to break the lateral body fall and accelerate the CoM in the direction of the stance hand. As a result, maintaining equivalent dynamical stability at the end of the handstand walking initiation phase, will prepare the body for the consecutive steps.

No significant difference was found in step parameters across the groups regardless of the positive relationship between step length and anteroposterior CoM displacement and velocity parameters. The lack of significant difference between the groups in step parameters could be explained by the similar anteroposterior CoM parameters between groups at hand-off. Based on the bipedal gait initiation model, previous studies showed that the CoP shift during the Preparation Phase serves to accelerate the CoM in the opposite direction [6–8]. The PSG increased the CoM velocity throughout the first 50 % of the Preparation Phase by performing a posterior CoP shift during the same phase, whereas the ASG reduced the CoM velocity by performing an opposite anteroposterior CoP trajectory during the first 75 % of the Preparation Phase. In addition, both groups performed a posterior CoP shift at the end of the Preparation Phase (75–100 %). As a result, both groups began the Execution Phase under similar conditions and, therefore, similar step parameters were observed.

5. Practical implications

Handstand walking is growing in popularity in gymnastics and other sports, e.g. CrossFit. Loss of balance and falls are often present in beginners during the first step of handstand walking; therefore, a good mechanical understanding of this motor task is necessary to improve the design of appropriate training protocols and reduce the risk of injury while improving handstand performance. On-field, the assessment of CoM and the CoP could also be provided by wearable sensors and pressure mats, respectively [25,26], as the use of such equipment is becoming more accessible during training. Therefore, coaches and athletes will be able to estimate the CoM and CoP behaviour during handstand walking training and compare it with the findings of the current paper. By doing so the coaches will be able to provide any necessary feedback to their athletes which can further improve their performance and ability to learn the task.

6. Conclusion

Our findings indicate that performing the first step in handstand walking initiation requires the CoM to be shifted forward and towards the stance hand. This is made possible by performing an appropriate CoP pattern. In this study, two main anteroposterior CoP shift strategies were observed and defined across the participants. Nevertheless, the difference in anteroposterior CoP trajectory during the Preparation Phase resulted in similar mediolateral behaviour and anteroposterior CoM velocity at hand-off and, therefore, whatever the preparatory strategy used, no differences in step characteristics were found. The understanding of the necessary CoP and CoM patterns required to perform handstand walking initiation could contribute to individualise training protocols oriented to improve learning and teaching of the task. The

novelty of this study and its results can be used in future research investigating handstand walking initiation. Future studies should consider and assess the contribution of muscle activity and how it controls CoP shift during the handstand walking initiation as well as examine the effect of the biomechanical constraints on the adaptation of the different strategies during handstand walking initiation; as this could help with further understanding of handstand walking.

Declaration of Competing Interest

None of the authors have financial or other conflicts of interest in regard to this research.

Acknowledgements

We would like to acknowledge the support of Cardiff Metropolitan University within the Erasmus Mobility program. The authors would like to thank Dr Hans VonLieres for his technical support, Mr Matteo Bonfiglio for his help with participant recruitment, Ms Amy Wynn, Ms Cecilia Bratta and Mr Paolo Brasiliano for their support during data collection and Ms Joanna Leigh for her spelling and grammar support.

References

- [1] Y. Brenière, M. Cuong Do, S. Bouisset, Are dynamic phenomena prior to stepping essential to walking? *J. Mot. Behav.* 19 (1) (1987) 62–76.
- [2] R. Lepers, Y. Breniere, The role of anticipatory postural adjustments and gravity in gait initiation, *Exp. Brain Res.* 107 (1) (1995) 118–124.
- [3] E. You, T. Caderby, A. Delafontaine, P. Fourcade, J.L. Honeine, Balance control during gait initiation: state-of-the-art and research perspectives, *World J. Orthop.* 8 (11) (2017) 815.
- [4] S.E. Halliday, D.A. Winter, J.S. Frank, A.E. Patla, F. Prince, The initiation of gait in young, elderly, and Parkinson's disease subjects, *Gait Posture* 8 (1) (1998) 8–14.
- [5] C. Lu, S.L.A. Huffmaster, J.C. Harvey, C.D. MacKinnon, Anticipatory postural adjustment patterns during gait initiation across the adult lifespan, *Gait Posture* 57 (2017) 182–187.
- [6] I.N. Lyon, B.L. Day, Control of frontal plane body motion in human stepping, *Exp. Brain Res.* 115 (2) (1997) 345–356.
- [7] L. Laudani, A. Casabona, V. Percivalle, A. Macaluso, Control of head stability during gait initiation in young and older women, *J. Electromyogr. Kinesiol.* 16 (2006) 603–610.
- [8] T. Caderby, E. You, N. Peyrot, M. Begon, G. Dalleau, Influence of gait speed on the control of mediolateral dynamic stability during gait initiation, *J. Biomech.* 47 (2) (2014) 417–423.
- [9] M. Sobera, B. Siedlecka, P. Piestrak, K. Sojka-Krawiec, B. Graczykowska, Maintaining body balance in extreme positions, *Biol. Sport* 24 (1) (2007) 81.
- [10] D.G. Kerwin, G. Trewartha, Strategies for maintaining a handstand in the anterior-posterior direction, *Med. Sci. Sports Exerc.* 33 (7) (2001) 1182–1188.
- [11] G. Gautier, R. Thouvenecq, D. Chollet, Visual and postural control of an arbitrary posture: the handstand, *J. Sports Sci.* 25 (11) (2007) 1271–1278.
- [12] S.M. Slobounov, K.M. Newell, Postural dynamics in upright and inverted stances, *J. Appl. Biomech.* 12 (2) (1996) 185–196.
- [13] P. Hedbávný, J. Sklenářková, D. Hupka, M. Kalichová, Balancing handstand on the floor, *Sci. Gymnastics J.* 5 (2013) 69–80.
- [14] G.M. Earhart, C. Mosiman, Coordination of handwalking in gymnasts: a comparison to bipedal walking, *Sci. Gymnastics J.* (ScGYM®) (2010) 41.
- [15] Fédération Internationale De Gymnastique, 2017–2020 Code of Point, Available at: http://www.fig-gymnastics.com/publicdir/rules/files/tra/TRA-CoP_2017-2020-e.pdf [Accessed: 20th April 2017], 2017.
- [16] P.E. Pidcoe, E.N. Burnet, Rehabilitation of an elite gymnast with a type II manubriosternal dislocation, *Phys. Ther.* 87 (4) (2007) 468–475.
- [17] D.A. Winter, F. Prince, J.S. Frank, C. Powell, K.F. Zabjek, Unified theory regarding A/P and M/L balance in quiet stance, *J. Neurophysiol.* 75 (6) (1996) 2334–2343.
- [18] M.P. Kadaba, H.K. Ramakrishnan, M.E. Wootten, Measurement of lower extremity kinematics during level walking, *J. Orthop. Res.* 8 (3) (1990) 383–392.
- [19] E.M. Gutierrez, A. Bartonek, Y. Haglund-Åkerlind, H. Saraste, Centre of mass motion during gait in persons with myelomeningocele, *Gait Posture* 18 (2003) 37–46.
- [20] L. Rum, G. Vannozzi, A. Macaluso, L. Laudani, Neuromechanical response of the upper body to unexpected perturbations during gait initiation in young and older adults, *Aging Clin. Exp. Res.* (2020), <https://doi.org/10.1007/s40520-020-01592-2>.
- [21] A. Chen, S. Selvaraj, V. Krishnan, S. Asgari, Accuracy and reliability of onset detection algorithms in gait initiation for healthy controls and participants with Parkinson's disease, *J. Appl. Biomech.* 1 (aop) (2019) 1–8.
- [22] M.P. Kadaba, H.K. Ramakrishnan, M.E. Wootten, J. Gainey, G. Gorton, G.V. B. Cochran, Repeatability of kinematic, kinetic, and electromyographic data in normal adult gait, *J. Orthop. Res.* 7 (1989) 849–860.

- [23] J.L. McGinley, R. Baker, R. Wolfe, M.E. Morris, The reliability of three-dimensional kinematic gait measurements: a systematic review, *Gait Posture* 29 (3) (2009) 360–369.
- [24] J. Cohen, *Statistical Power Analysis for the Behavioral Sciences*, Revised ed., Academic Press, New York, 1977.
- [25] V. Camomilla, E. Bergamini, S. Fantozzi, G. Vannozzi, Trends supporting the in-field use of wearable inertial sensors for sport performance evaluation: a systematic review, *Sensors* 18 (3) (2018) 873.
- [26] J. Goetschius, M.A. Feger, J. Hertel, J.M. Hart, Validating center-of-pressure balance measurements using the MatScan® pressure mat, *J. Sport Rehabil.* 27 (1) (2018).